

Absolute maximization forces a near-maximal base

MathlibAnnex.Satellite.absoluteMaximizer_base_nearMax

theorem

A dominant determinant term prevents a large deficit in the base frame.

Statement

Let $n \in \mathbb{N}$ and use reference coordinates $V = \mathbb{R}^n$ with the sup norm. Let $p : V \rightarrow \mathbb{R}$ be a continuous seminorm with specified constants $a_p, b_p > 0$ such that

$$a_p \|x\|_\infty \leq p(x) \leq b_p \|x\|_\infty \quad (x \in V).$$

Thus p is a norm. We use p for the model distance and $\|\cdot\|_\infty$ for reference-coordinate estimates; the two norms are allowed to coincide. A p -dual contraction is a continuous real linear functional r satisfying $|r(x)| \leq p(x)$ for every x . Let $e_1, \dots, e_n$ be the standard coordinate vectors. For an ordered family $F = (\ell_i)_{i=1}^n$ of continuous real linear functionals on V , put

$$M_F = (\ell_i(e_j))_{i,j}, \quad T_F x = (\ell_i(x))_i = M_F x, \quad \mathcal{U}_p = \{F : \forall i, x, |\ell_i(x)| \leq p(x)\}.$$

For a real number η , define the attained maximum and near-maximal set by

$$D_p = \max_{F \in \mathcal{U}_p} |\det M_F| > 0, \quad \mathcal{N}_{p,\eta} = \{F \in \mathcal{U}_p : D_p - \eta \leq |\det M_F|\}.$$

Admissibility is measured with p ; the matrices are written in the fixed reference coordinates.

Let J be a finite index set, possibly empty; choose coefficients $c_a \in V$ for $a \in J$. A configuration is a pair $Q = (F, s)$, where F is a frame and $s = (s_a)_{a \in J}$ is a family of additional continuous real linear functionals, called satellites. Write

$$\Omega_p^J = \{(F, s) : F \in \mathcal{U}_p, \forall a, x, |s_a(x)| \leq p(x)\}.$$

For any real weight w , let $M_F[i \leftarrow r]$ replace row i by $(r(e_j))_j$ and set

$$\Phi_{w,c}(F, s) = w \det M_F + \sum_{a \in J} \sum_{i=1}^n (c_a)_i \det M_F[i \leftarrow s_a].$$

An absolute maximizer means a single $Q \in \Omega_p^J$ with $|\Phi_{w,c}(Q')| \leq |\Phi_{w,c}(Q)|$ for every $Q' \in \Omega_p^J$. It need not maximize the signed polynomial. Set

$$S_p(c) = D_p \sum_{a \in J} \sum_{i=1}^n |(c_a)_i|.$$

Suppose $\eta \geq 0$, $w > 0$, $S_p(c) < w\eta$, and $Q = (F, s)$ is an admissible absolute maximizer. Then $F \in \mathcal{N}_{p,\eta}$.

Assumptions

The stated positive comparison constants for p , finiteness of J , nonnegative deficit, positive weight, strict budget inequality, admissibility of Q and its global absolute maximality are required. There is no $\eta < D_p$ assumption in this theorem.

Conclusion

The base frame has $|\det M_F| \geq D_p - \eta$ while remaining p -dual contractive. Invertibility requires the separate additional condition $\eta < D_p$.

Proof route

Bound the satellite sum and compare against an attained maximum frame with zero satellites.

▼ Proof steps

1. **Bound every satellite contribution.** For an admissible Q , replacing a row by any s_a leaves a p -dual contraction frame. Therefore each replacement determinant has absolute value at most D_p . Applying the finite satellite budget estimate to precisely these contraction rows gives

$$\left| \sum_{a,i} (c_a)_i \det M_F[i \leftarrow s_a] \right| \leq D_p \sum_{a,i} |(c_a)_i| = S_p(c).$$

Since $w \geq 0$, the absolute polynomial upper bound yields $|\Phi_{w,c}(Q)| \leq w|\det M_F| + S_p(c)$.

2. **Use a benchmark in the same domain.** Choose a p -dual frame F_* attaining D_p , and set all its satellites to zero. This comparison configuration is admissible. Its polynomial has absolute value wD_p . Absolute maximality of the given Q gives

$$wD_p = |\Phi_{w,c}(F_*, 0)| \leq |\Phi_{w,c}(Q)|.$$

This comparison uses the same weight and coefficients, without choosing a sign for the determinant.

3. **Exclude a deficit larger than η .** If $|\det M_F| < D_p - \eta$, positivity of w and $S_p(c) < w\eta$ imply

$$|\Phi_{w,c}(Q)| \leq w|\det M_F| + S_p(c) < w(D_p - \eta) + w\eta = wD_p,$$

contradicting the benchmark. Thus $D_p - \eta \leq |\det M_F|$. Together with $F \in \mathcal{U}_p$ this is near-maximal membership, the output of the benchmark-to-near-maximal lemma.

The budget is defined by the determinant-scaled coefficient sum. For later applications with $\eta > 0$, a weight dominating that budget chooses $w = S_p(c)/\eta + 1$, giving $w > 0$ and $w\eta = S_p(c) + \eta > S_p(c)$. This weight choice is a contextual result with the stronger positive-deficit input, not an additional assumption on this theorem.

Main citations

- the finite satellite budget estimate — `MathlibAnnex.Satellite.abs_satelliteTerms_le_budget`
- the absolute polynomial upper bound — `MathlibAnnex.Satellite.abs_satellitePolynomial_le`

- the benchmark-to-near-maximal lemma — `MathlibAnnex.Satellite.base_mem_nearMax_of_benchmark_le`
- the determinant-scaled coefficient sum — `MathlibAnnex.Satellite.satelliteBudget`
- a weight dominating that budget — `MathlibAnnex.Satellite.exists_weight_dominating_budget`
- Exact declaration and proof — `MathlibAnnex.Satellite.absoluteMaximizer_base_nearMax`

Lean source signature (exact)

```
theorem absoluteMaximizer_base_nearMax {n : ℕ} {J : Type u}
  [Fintype J] (M : NormModel n) {η weight : ℝ}
  (hη0 : 0 ≤ η) (hweight : 0 < weight)
  (coeff : J → Coord n)
  (hgap : satelliteBudget M coeff < weight * η)
  {C : SatelliteConfiguration n J}
  (hC : C ∈ satelliteConfigurationSet M J)
  (hmax : ∀ D ∈ satelliteConfigurationSet M J,
    |configurationPolynomial weight coeff D| ≤
    |configurationPolynomial weight coeff C|) :
  C.1 ∈ nearMaxFrames M η
```

The proof arguments `hη0`, `hweight`, `hgap`, `hC` and `hmax` are assumptions. The final membership is the conclusion.

In the source	Mathematical meaning
<code>Coord n</code>	The reference space $V = \mathbb{R}^n$ with its sup norm; <code>Coord n</code> abbreviates $\text{Fin } n \rightarrow \mathbb{R}$. Lean uses indices $0, \dots, n-1$ and the formulas use $1, \dots, n$.
<code>M : NormModel n</code> ; <code>M.p</code>	The input <code>M</code> contains the continuous seminorm p and constants $a_p, b_p > 0$ with $a_p \ x\ _\infty \leq p(x) \leq b_p \ x\ _\infty$ for every x . <code>M.p x</code> is the scalar $p(x)$.
<code>[Fintype J]</code>	The index set J is finite and may be empty; this permits the finite sum over all $a \in J$.
<code>DualFrame n</code> ; <code>dualFrameMatrix B</code> ; <code>frameDet B</code>	An ordered family $F = (\ell_i)$ of continuous real linear functionals, its matrix $M_F = (\ell_i(e_j))_{i,j}$ in the standard basis, and its determinant $\det M_F$. No independence is imposed by the family type.
<code>detMax M</code> ; <code>nearMaxFrames M η</code>	$D_p = \max_{F \in \mathcal{U}_p} \det M_F $ and $\mathcal{N}_{p,\eta} = \{F \in \mathcal{U}_p : D_p - \eta \leq \det M_F \}$, where $\mathcal{U}_p = \{(\ell_i) : \forall i, x, \ell_i(x) \leq p(x)\}$.
<code>SatelliteConfiguration n J</code> ; <code>C.1</code> ; <code>C.2 a</code>	A pair $Q = (F, s)$, called <code>C</code> . Its first component <code>C.1</code> is the base frame F ; <code>C.2 a</code> is the satellite $s_a : V \rightarrow \mathbb{R}$, continuous and real linear.

In the source	Mathematical meaning
weight; coeff a; coeff a i	The real weight w , the coefficient vector $c_a \in V$, and its i th coordinate $(c_a)_i$. These coefficients are fixed before maximization.
hC : C ∈ satelliteConfigurationSet M J	The given configuration lies in Ω_p^J : $ \ell_i(x) \leq p(x)$ and $ s_a(x) \leq p(x)$ for every i, a, x .
configurationPolynomial weight coeff C	The scalar $\Phi_{w,c}(Q) = w \det M_F + \sum_{a \in J} \sum_i (c_a)_i \det M_F[i \leftarrow s_a]$, where the replaced row is $(s_a(e_j))_j$.
hmax : ∀ D ∈ satelliteConfigurationSet M J, ...	For every admissible comparison pair $Q' \in \Omega_p^J$, $ \Phi_{w,c}(Q') \leq \Phi_{w,c}(Q) $. D names that pair, not the number D_p . This is a hypothesis about absolute values.
hη0; hweight; hgap	The three hypotheses $0 \leq \eta$, $0 < w$, and $S_p(c) < w\eta$, respectively.

In the source	Mathematical meaning
satelliteBudget M coeff	The scalar $S_p(c) = D_p \sum_{a \in J} \sum_i (c_a)_i $.
C.1 ∈ nearMaxFrames M η	The conclusion is $F \in \mathcal{U}_p$ and $D_p - \eta \leq \det M_F $, for the base of the same pair C.

Here $\rightarrow L[\mathbb{R}]$ means continuous real linear, $\forall$ means “for every”, and $\in$ is membership. Dot notation selects a field or component; successive arguments denote function application.

▼ Relevant surrounding context (separate exact excerpts)

Exact source lines 13–15:

```
private abbrev Coord (n : ℕ) := Fin n → ℝ
private abbrev NormModel (n : ℕ) := EquivalentSeminorm (Coord n)
private abbrev DualFrame (n : ℕ) := DeterminantFrame.Frame n (Coord n)
```

Exact source lines 53–53:

```
private abbrev detMax {n : ℕ} (M : NormModel n) := DeterminantFrame.determinantMaximum (modelBasis M)
```

Exact source lines 55–55:

```
private abbrev nearMaxFrames {n : ℕ} (M : NormModel n) (η : ℝ) := {B : DualFrame n | B ∈ dualFrameSet M
  ∧ detMax M - η ≤ |frameDet B|}
```

Full surrounding source. These excerpts are separate from the declaration above.

Surrounding assumptions and aliases: exact source, lines 7–55. The full original context is retained with the source evidence.

▼ **Exact content identity**

Declaration: MathlibAnnex.Satellite.absoluteMaximizer_base_nearMax

Accepted content SHA-256: 1510ad9ef49cb88b88be7c6bd8311e1a66ff84728cb7c042fbf5b4acf64f4afc

Accepted source guide SHA-256: 3901d0f28fe6638210e9f1708754b0710dedc6d7f1203d12e53acc9bd1a2630d

Source commit: 437e6e46228bbb8e91211ebded349d7a30020e73